I2C Communication

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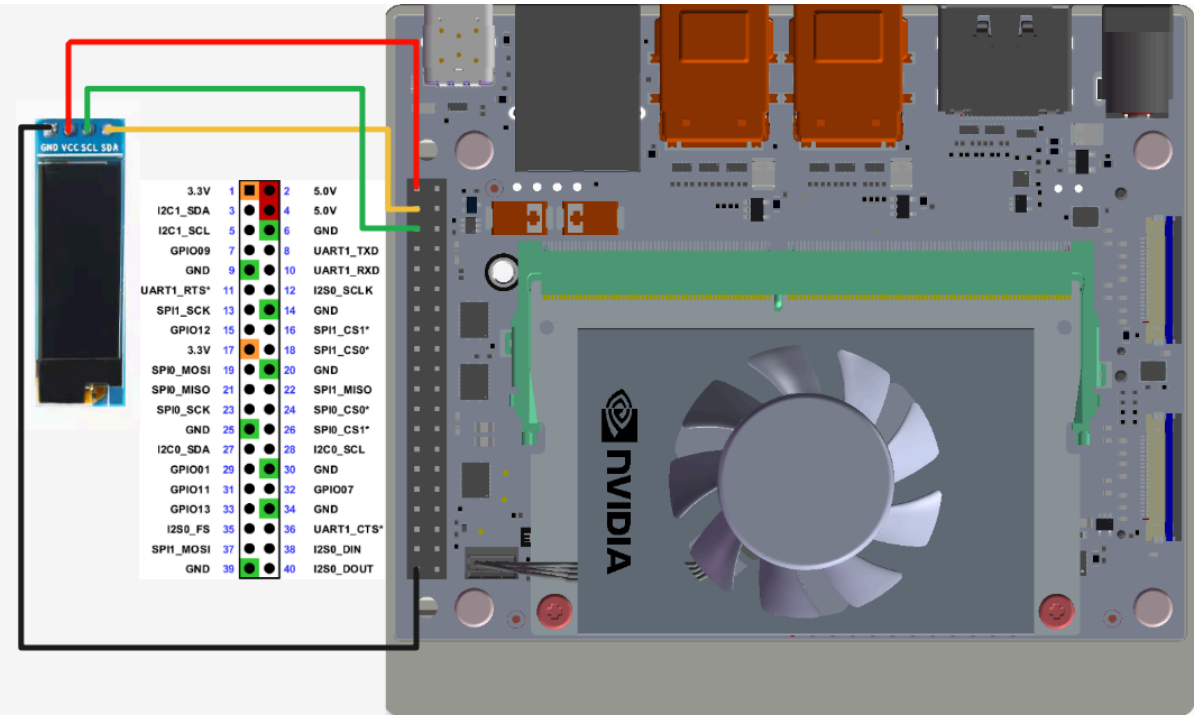
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> **pip Installation Note:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; a virtual environment usually does not need this parameter, but this tutorial keeps it to keep the commands consistent.

1. GPIO Pin Diagram

Use a 0.91-inch OLED to test the I2C communication function, and connect it according to the wiring below:

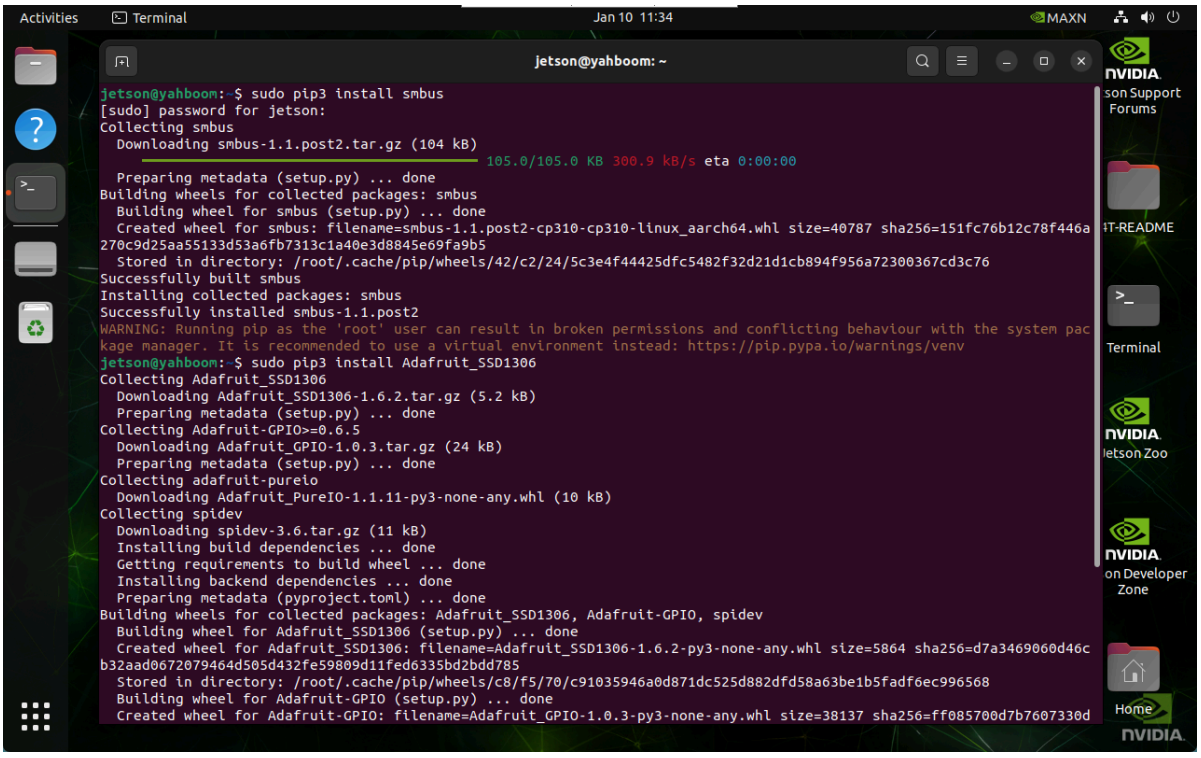
Note: Do not connect the pins incorrectly or cause a pin short circuit; a mistake may damage the mainboard hardware!



2. I2C Test

2.1. Install Dependencies

```
sudo pip3 install smbus --break-system-packages
sudo pip3 install Adafruit_SSD1306 --break-system-packages
```



2.2. I2C Devices

During normal development, you need to find the device bus to which the I2C device is mounted and the device address.

2.2.1. Query the I2C Bus

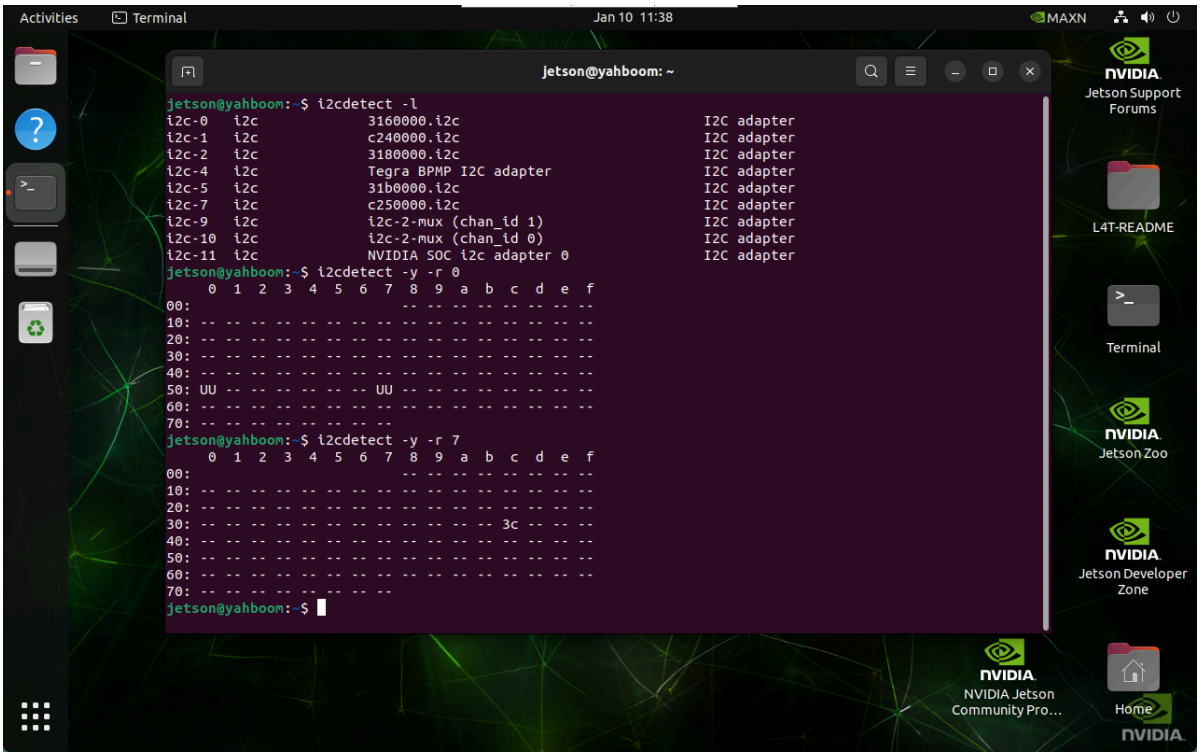
Enter the following command in the terminal to list all buses of the device:

```
i2cdetect -l
```

2.2.2. Query the I2C Device

Enter the following command in the terminal to list the I2C devices under the specified bus: the I2C address corresponding to the oled is 0x3c

```
i2cdetect -y -r *
```



2.3. Run the Program

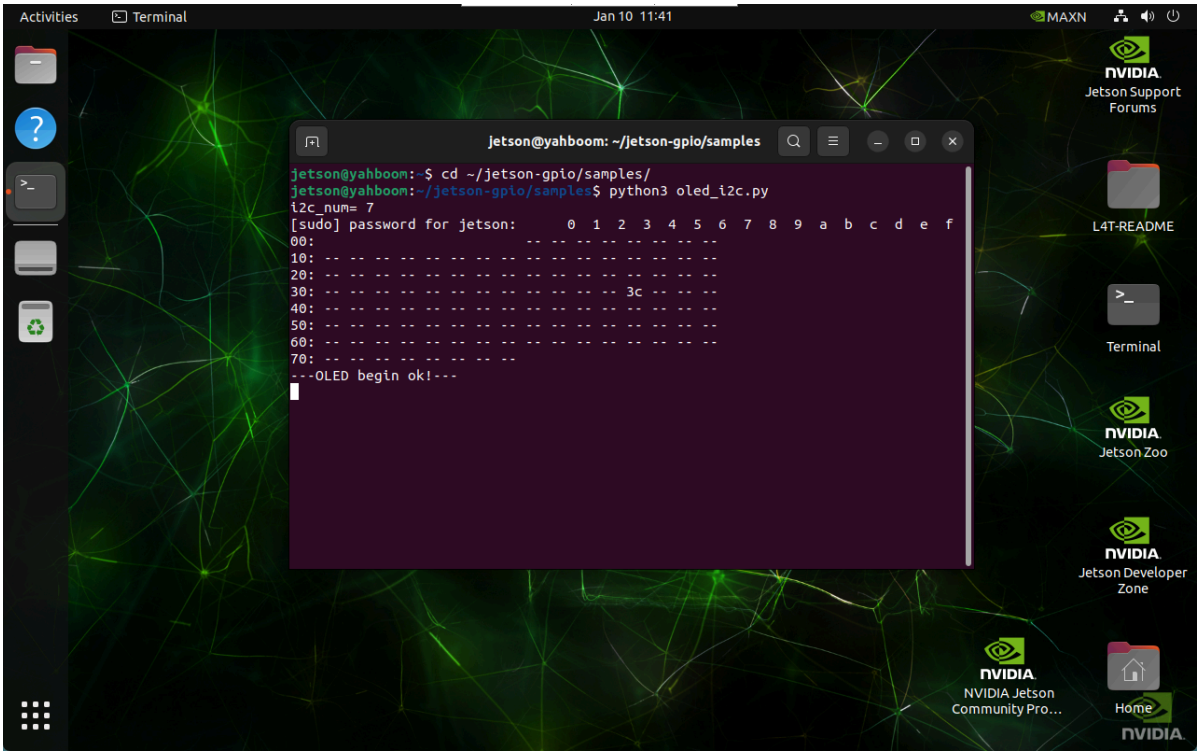
`oled_i2c.py` is not included in the `jetson-gpio` library:

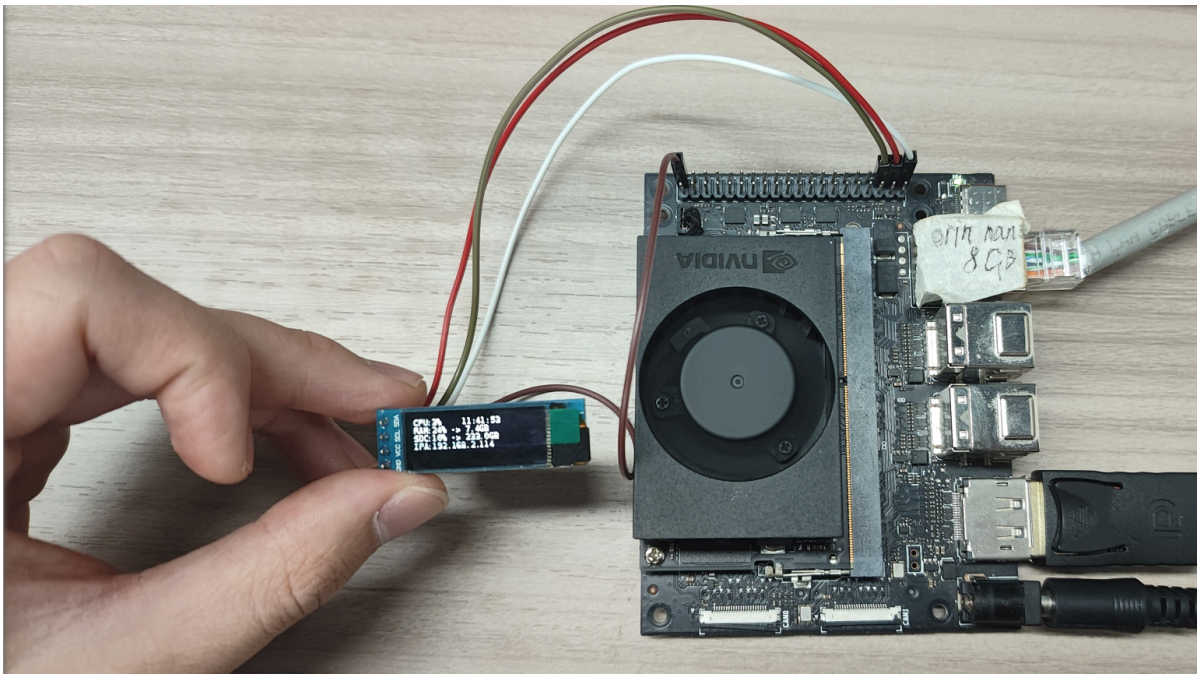
```
cd ~/jetson-gpio/samples/
```

```
python3 oled_i2c.py
```

3. Experiment Effect

After the program is started, the OLED displays system information such as the system CPU usage, the system time, and the memory usage:




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